

QSPRpred: a Flexible Open-Source Quantitative Structure-Property Relationship Modelling Tool

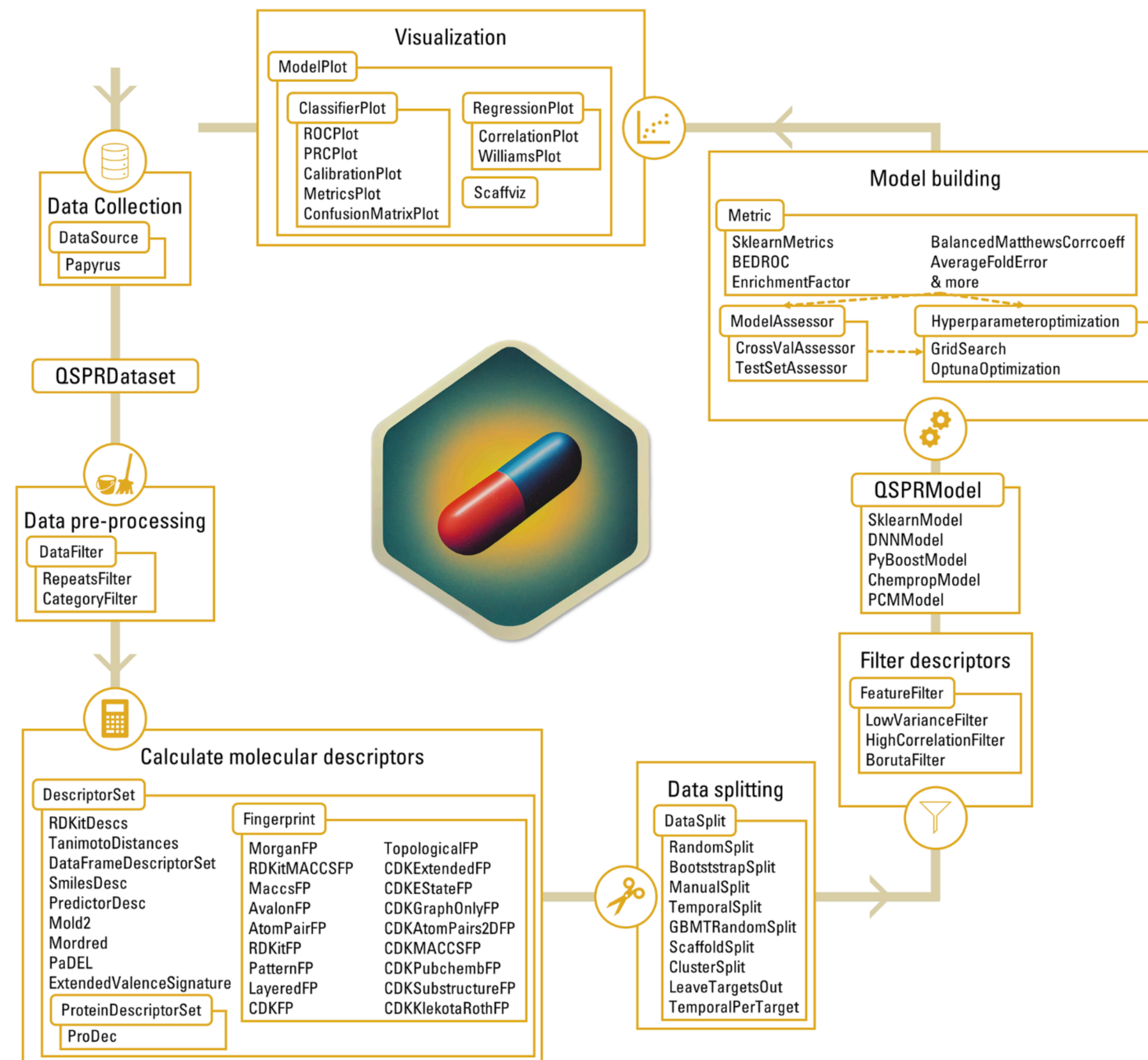
Adding Dimensionality Reduction Plots, Biased Splitting, and Fixing bugs in CLI.

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Introduction

- Open-source Python package.
- Designed for QSPR/QSAR modelling.
- Covers the **entire workflow** from data preparation to model deployment.



Data Processing

```
from qsprpred.data import QSPRDataset
import os

os.makedirs("tutorial_output/data", exist_ok=True)

dataset = QSPRDataset(
    df=df,
    store_dir="tutorial_output/data",
    name="QuickStartDataset",
    target_props=[{"name": "pchembl_value_Mean", "task": "REGRESSION"}],
    random_state=42
)

dataset.getDF()
```

Load Data

Failed to find the pandas get_adjustment() function to patch
Failed to patch pandas – PandasTools will have limited functionality

	SMILES	pchembl_value_Mean	Year	QSPRID	pchembl_value_Mean_original
QSPRID					
QuickStartDataset_0000	Cc1nn(-c2cc(NC(=O)CCN(C)C)nc(-c3ccc(C)o3)n2)c(...	8.68	2008.0	QuickStartDataset_0000	8.68
QuickStartDataset_0001	Nc1c(C(=O)Nc2ccc([N+](=O)[O-])cc2)sc2c1cc1CCCC...	4.82	2010.0	QuickStartDataset_0001	4.82
QuickStartDataset_0002	O=C(Nc1nc2ncccc2n2c(=O)n(-c3ccccc3)nc12)c1ccccc1	5.65	2009.0	QuickStartDataset_0002	5.65
QuickStartDataset_0003	CNC(=O)C12CC1C(n1cnc3c1nc(C#CCCCC(=O)OC)nc3NC...	5.45	2009.0	QuickStartDataset_0003	5.45
QuickStartDataset_0004	CCCN1c(=O)c2c(nc3cc(OC)ccn32)n(CCCNC(=O)c2ccc(...	5.20	2019.0	QuickStartDataset_0004	5.20
...	...	...	...	...	...
QuickStartDataset_4077	CNc1ncc(C(=O)NCc2ccc(OC)cc2)c2nc(-c3ccco3)nn12	7.09	2018.0	QuickStartDataset_4077	7.09
QuickStartDataset_4078	Nc1nc(-c2ccco2)c2ncn(C(=O)NCCc3ccccc3)c2n1	8.22	2008.0	QuickStartDataset_4078	8.22
QuickStartDataset_4079	Nc1nc(Nc2ccc(F)cc2)nc(CSc2nnc(N)s2)n1	4.89	2010.0	QuickStartDataset_4079	4.89
QuickStartDataset_4080	CCCOc1ccc(C=Cc2cc3c(c(=O)n(C)c(=O)n3C)n2C)cc1	6.51	2013.0	QuickStartDataset_4080	6.51
QuickStartDataset_4081	CCOC(=O)c1cnc(NCC(C)C)n2nc(-c3ccco3)nc12	7.35	2014.0	QuickStartDataset_4081	7.35

Data Processing

```
from qsprpred.data.descriptors.fingerprints import MorganFP
from qsprpred.data import RandomSplit

# Specifiy random split for creating the train (80%) and test set (20%)
rand_split = RandomSplit(test_fraction=0.2, dataset=dataset)

# calculate compound features and split dataset into train and test
dataset.prepareDataset(
    split=rand_split,
    feature_calculators=[MorganFP(radius=3, nBits=2048)],
)

print(f"Number of samples in train set: {len(dataset.y)}")
print(f"Number of samples in test set: {len(dataset.y_ind)}")

dataset.save()
```

Split Dataset*

* We have
implemented a set of
novel Biased Data
Splitting Methods

Number of samples in train set: 3265
Number of samples in test set: 817

Model Training

```
from qsprpred.models import SklearnModel
from sklearn.neighbors import KNeighborsRegressor
from qsprpred.models import CrossValAssessor, TestSetAssessor
```

```
os.makedirs("tutorial_output/models", exist_ok=True)
```

```
# This is an SKlearn model, so we will initialize it with the SklearnModel class
```

```
model = SklearnModel(
    base_dir="tutorial_output/models",
    alg=KNeighborsRegressor,
    name="QuickStartModel"
)
```

```
# We can now assess the model performance on the training set using cross validation
CrossValAssessor("r2")(model, dataset)
```

```
# and on the test set
```

```
TestSetAssessor("r2")(model, dataset)
```

```
# Finally, we need to fit the model on the complete dataset if we want to use it further
# This will save the fitted model and metadata to disk
model.fitDataset(dataset)
```

```
# We can optionally save the model and metadata to disk explicitly as well
```

```
_ = model.save()
```

Select Model

Test & Validate
Model

Save the
Trained Model

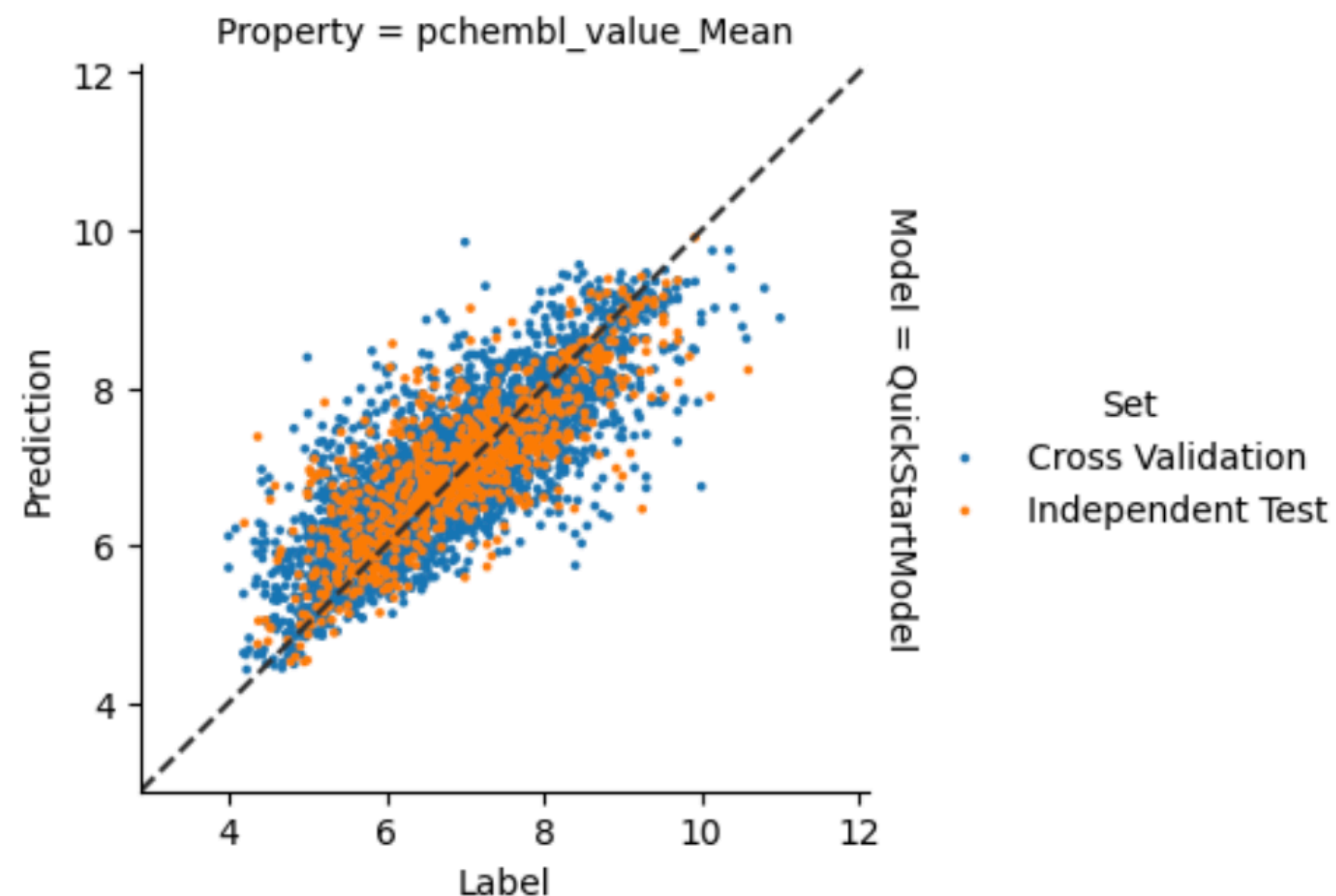
Data Visualization

```
from qsprpred.plotting.regression import CorrelationPlot  
plot = CorrelationPlot([model])  
axes, summary = plot.make(save=False, show=True)
```

Plotting Utils*

* We have added
PCA, UMAP, and
tSNE to these
plotting utils.

QuickStartModel



Command Line Interface

```
>>> python -m qsprpred.data_CLI
```

```
    -i ./tutorial_data/AR_LIGANDS_pivot.tsv    Input
```

```
        -o ./tutorial_output/data    Output
```

```
            -pr P29274    Papyrus
```

```
                -r REG    Regression
```

```
            -sp random    Split fraction
```

```
                -sf 0.15    Splitting
```

```
            -fe Morgan    Featurizer
```


Chemical Space Visualization

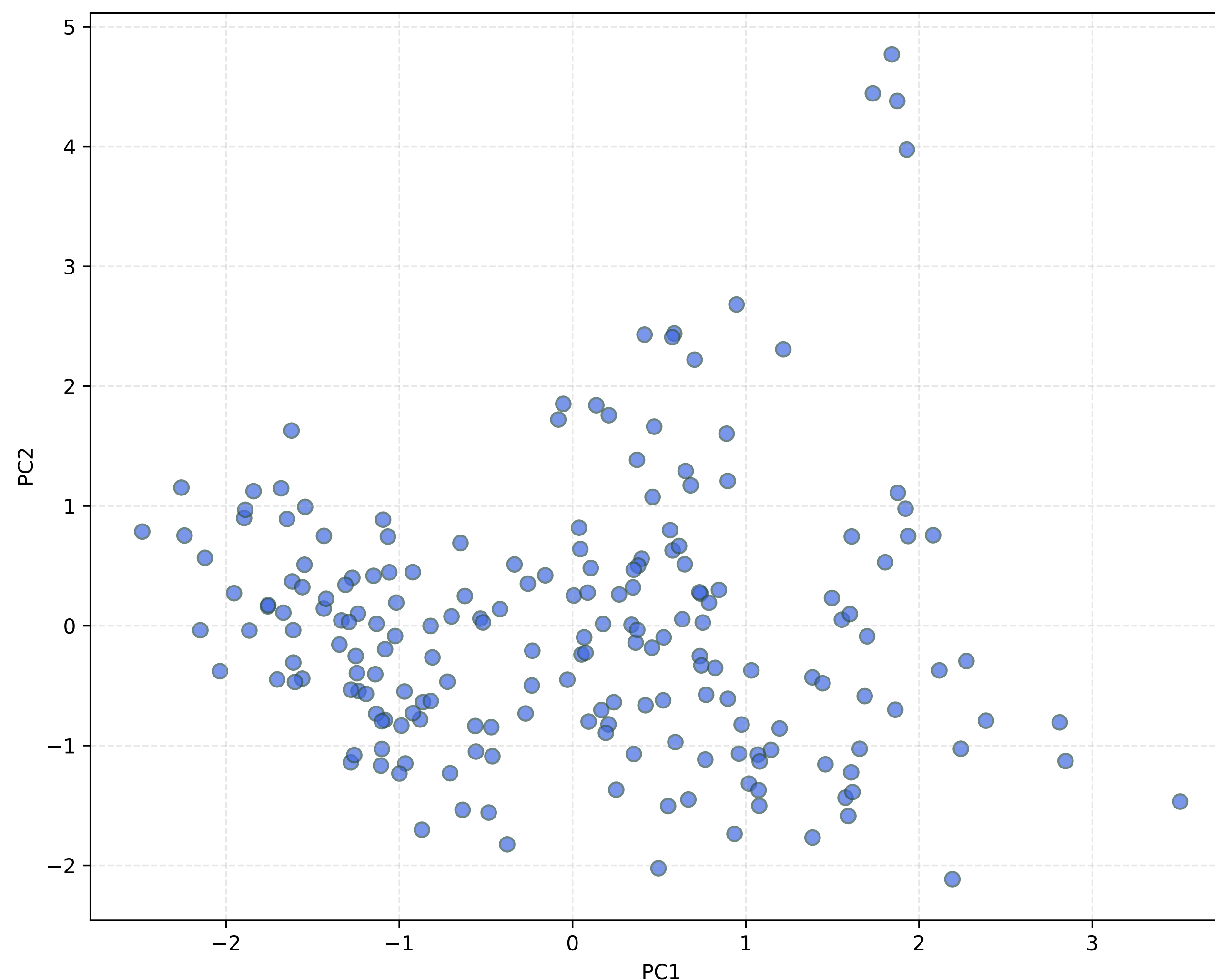


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```
>>> from qsprpred.plotting.chemicalspace import PCAPlot
```



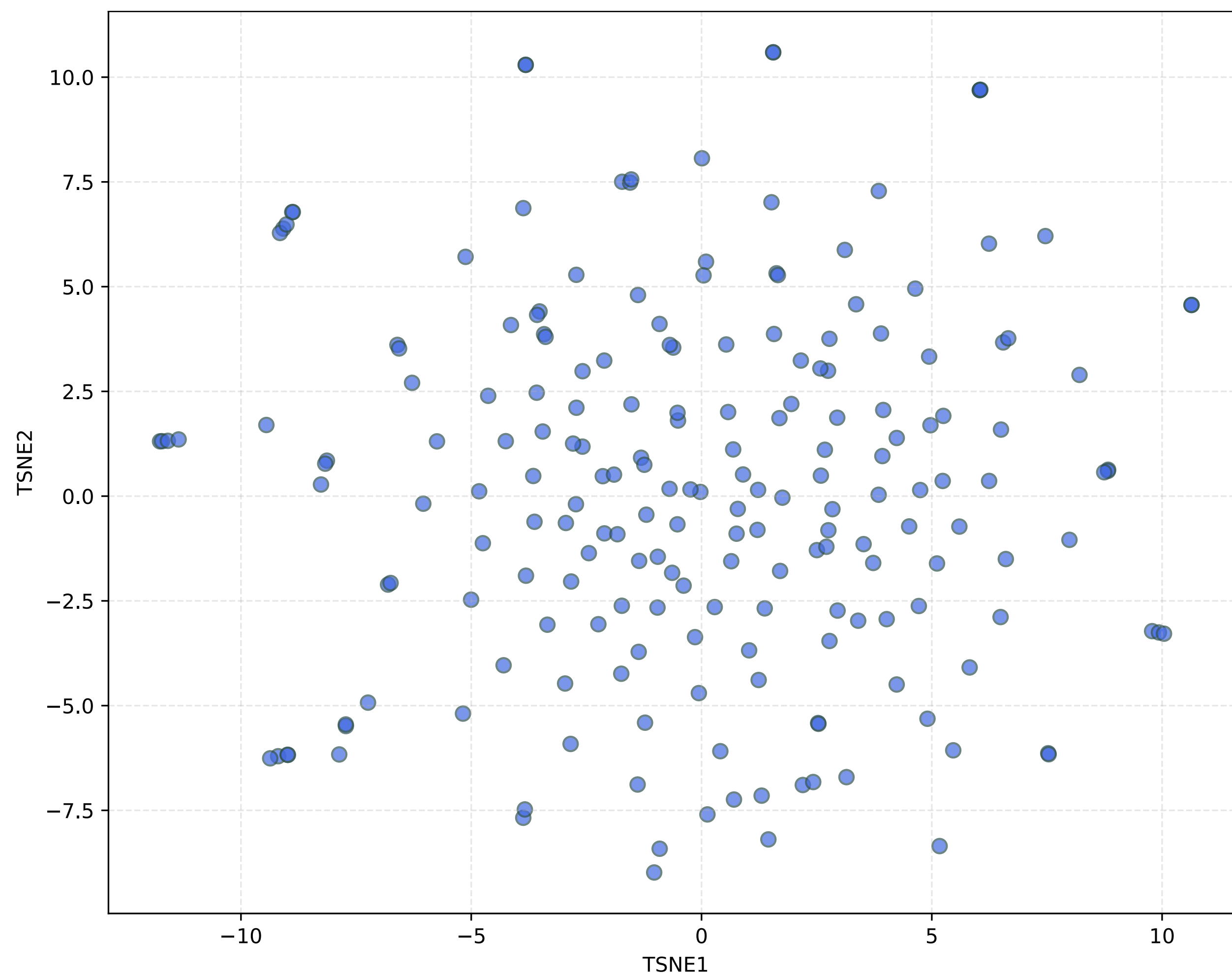
Chemical Space Visualization



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```
>>> from qsprpred.plotting.chemicalspace import TSNEPlot
```



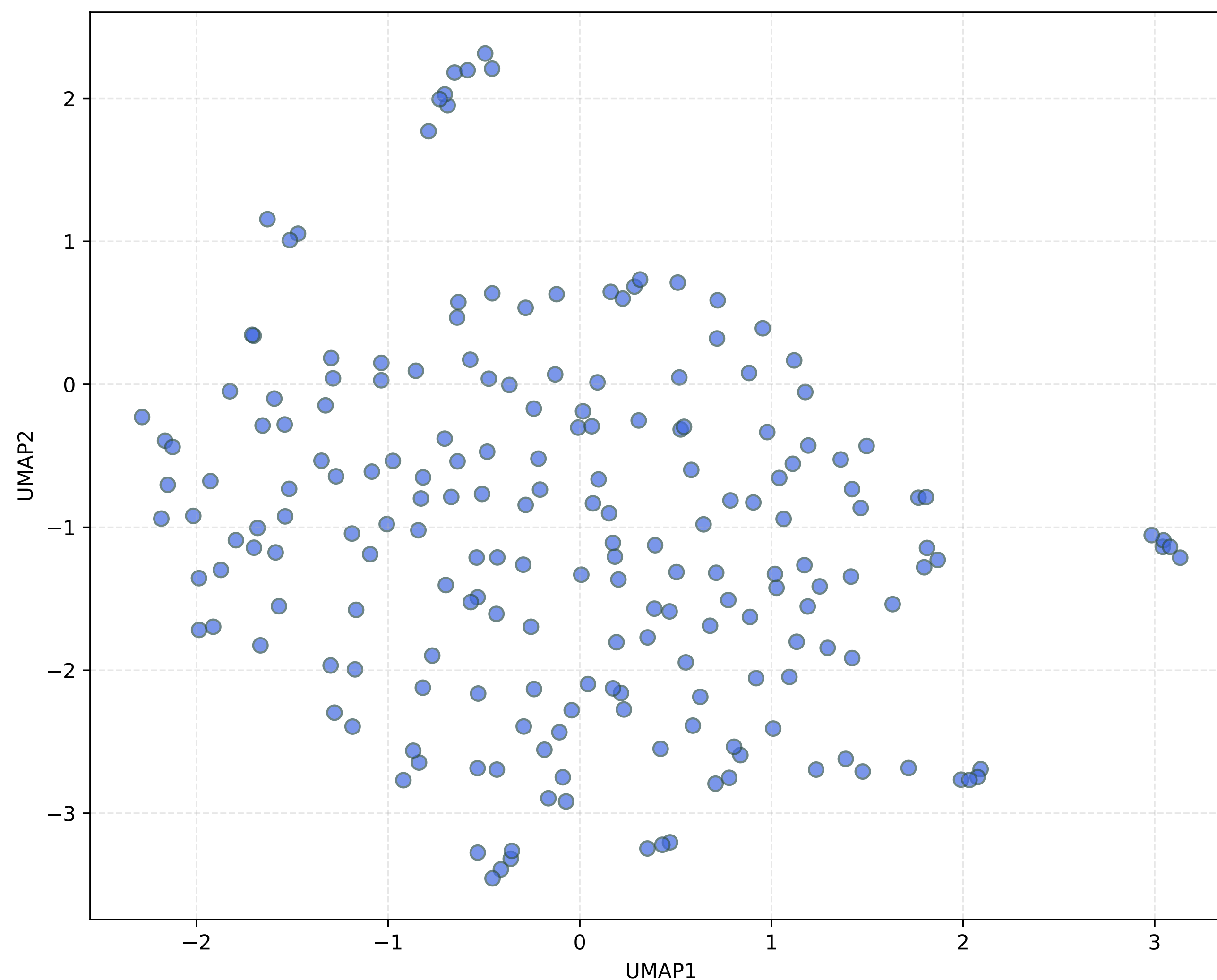
Chemical Space Visualization



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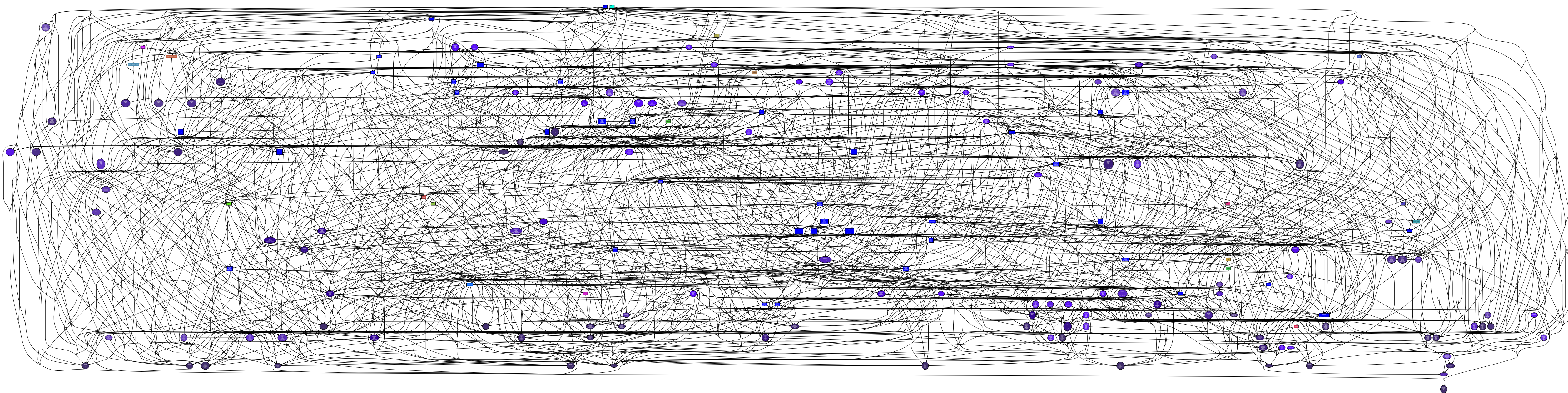
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```
>>> from qsprpred.plotting.chemicalspace import UMAPPlot
```



Package Dependency Plot

```
>>> pydeps qsprpred --max-bacon=2 --rankdir TB --cluster
```



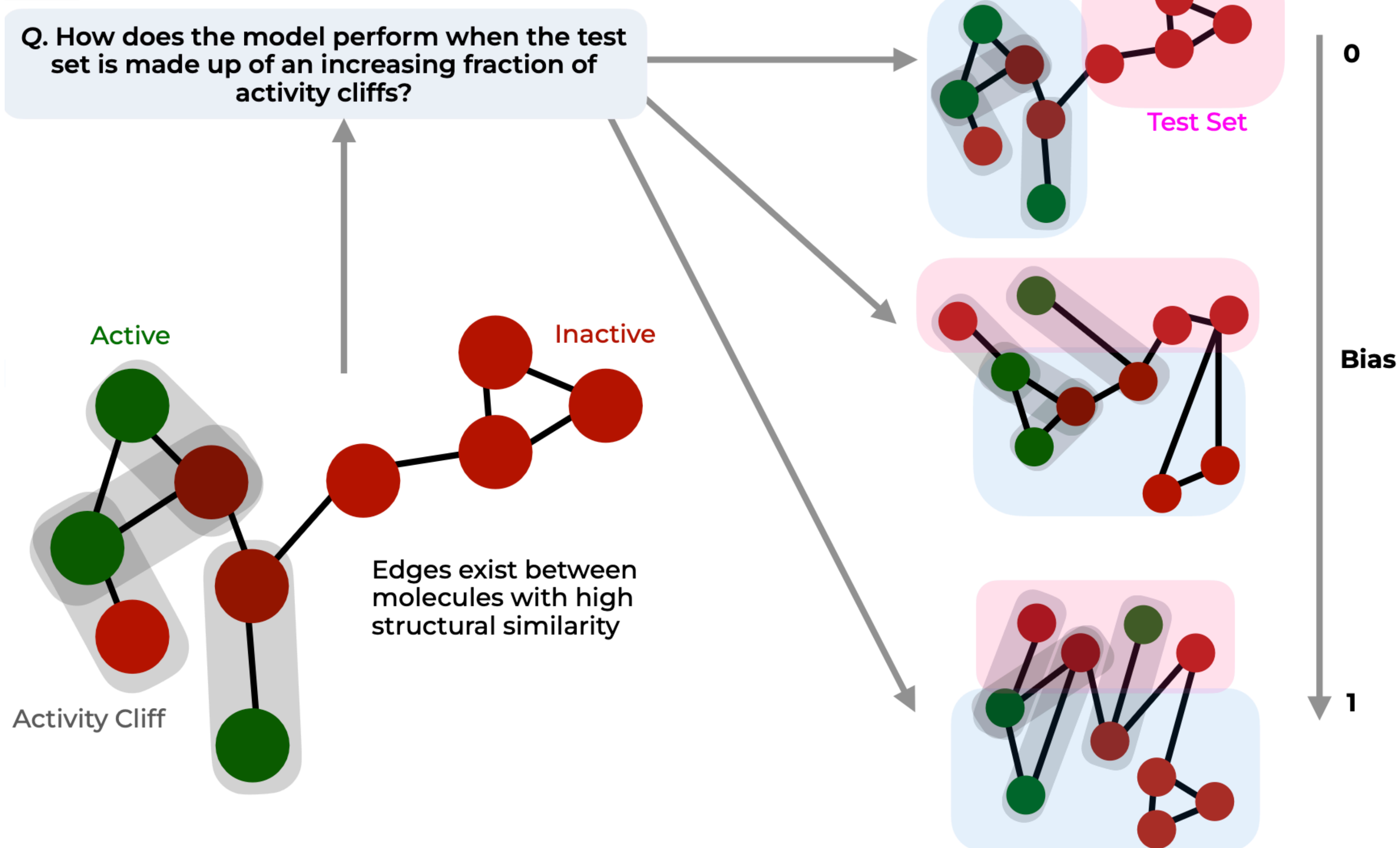
It's complicated ...

Biased Splits

pip install biased-split
<https://github.com/Manas02/biased-split>



Activity Cliff Splitting



Biased Splits

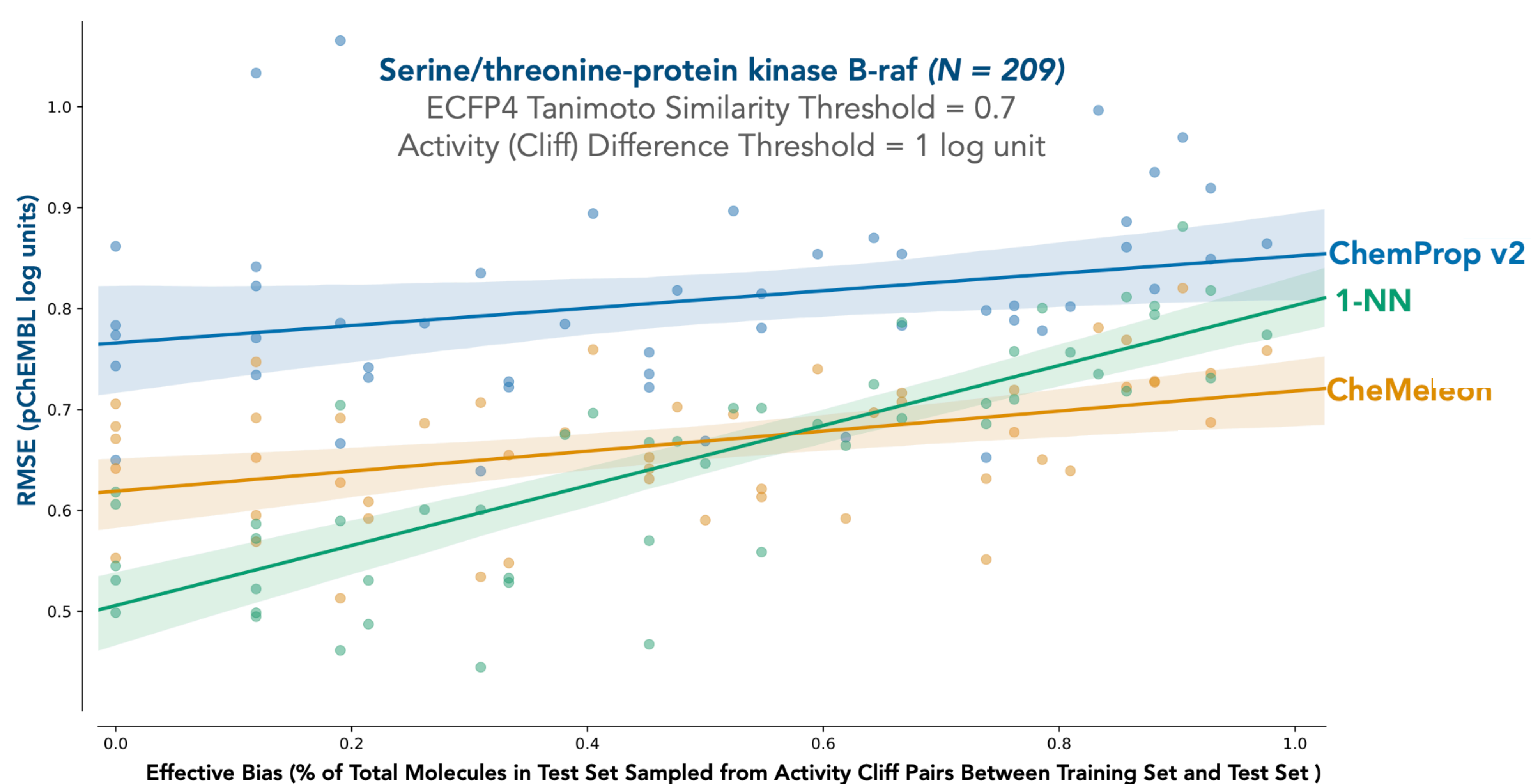
pip install biased-split
<https://github.com/Manas02/biased-split>



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Activity Cliff Splitting

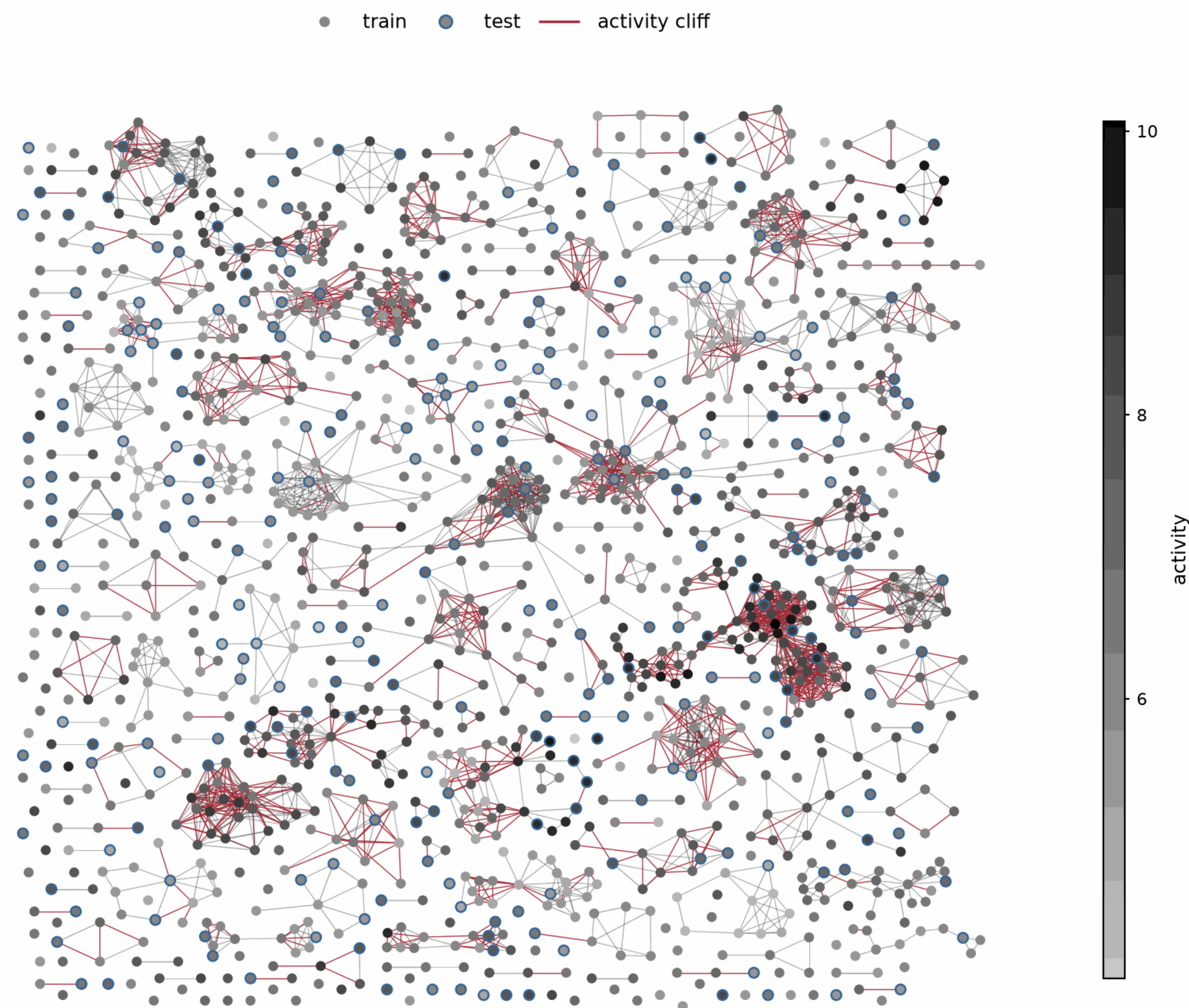


Biased Splits

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kNN Failure Split



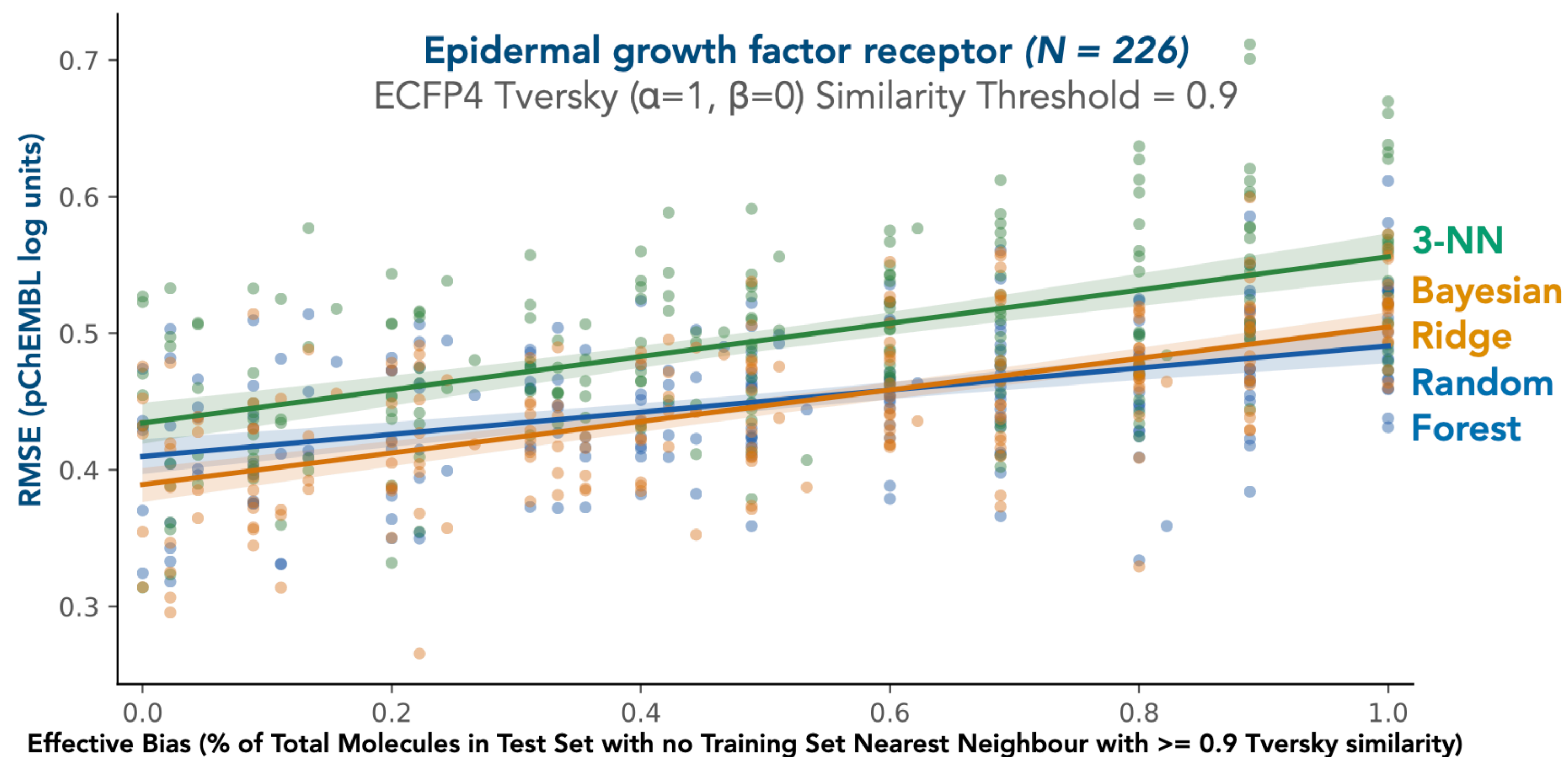
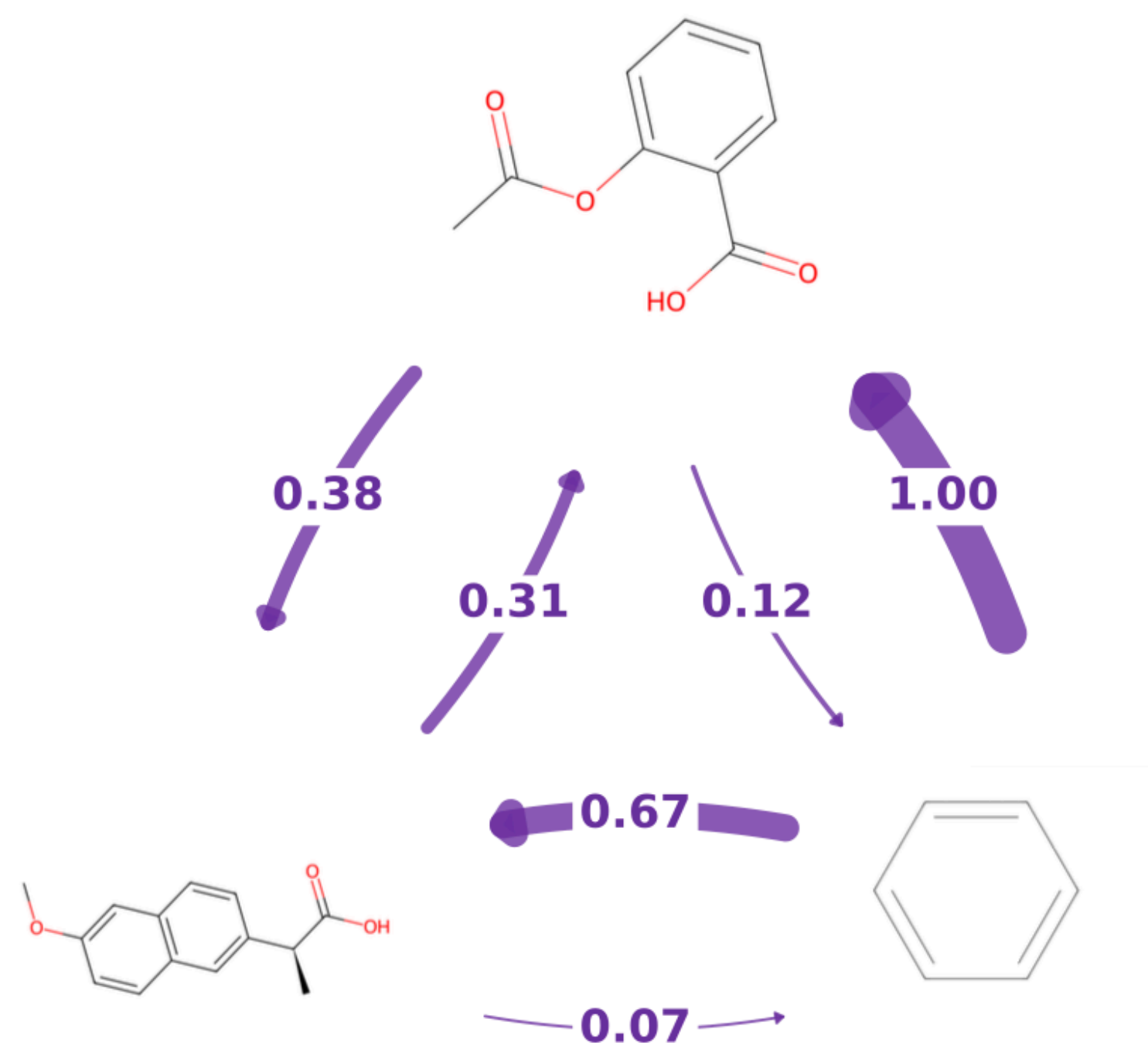
1348 molecules, 3276 edges. 2048bit ECFP4, tanimoto ≥ 0.7 . intended bias 0.00, effective bias 0.10

Biased Splits

pip install biased-split
<https://github.com/Manas02/biased-split>

Substructure Distance Split

Tversky Similarity using $\alpha=1, \beta=0$



Biased Splits

pip install biased-split
<https://github.com/Manas02/biased-split>



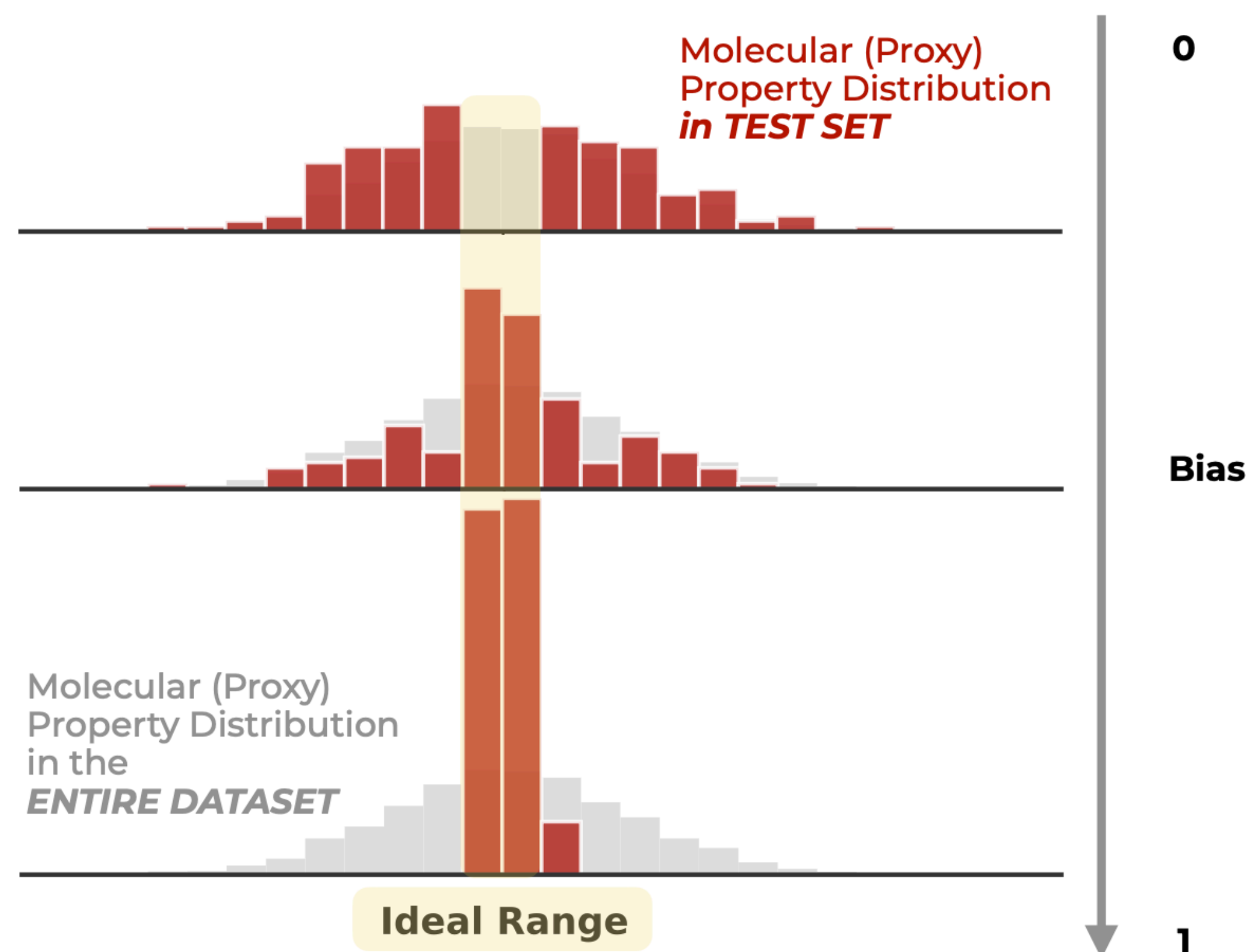
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Proxy Sorted Split

Q. How does the model perform when the test set is made up of an increasing fraction of compounds from the ideal proxy property range ?



Biased Splits — Test

```
manasmahale@Manas-MacBook-Pro:~/Desktop/QSPRpred/testing/test_pytest 81% 22/06/26 08:03
collected 10 items

../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testActivityCliffPostHocVerification PASSED [ 10%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testActivityCliffSplit PASSED [ 20%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testActivityCliffSplitSingleBias PASSED [ 30%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testActivityCliffSplitWithRepeats PASSED [ 40%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testKNNFailureSplit PASSED [ 50%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testKNNFailureSplitSingleBias PASSED [ 60%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testReproducibility PASSED [ 70%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testSubstructureDistancePostHocVerification PASSED [ 80%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testSubstructureDistanceSplit PASSED [ 90%]
../qsprpred/data/sampling/biased_split/tests.py::TestBiasedSplits::testSubstructureDistanceSplitSingleBias PASSED [100%]

===== warnings summary =====
qsprpred/data/sampling/biased_split/tests.py: 20 warnings
  /Users/manasmahale/Desktop/QSPRpred/qsprpred/data/tables/pnds.py:326: Pandas4Warning: The 'verify_integrity' keyword in DataFrame.set_index is deprecated and will be removed in a future version. Directly check the result.index.is_unique instead.
    self.df.set_index(self.idProp, inplace=True, verify_integrity=False, drop=False)

qsprpred/data/sampling/biased_split/tests.py: 10 warnings
  /Users/manasmahale/Desktop/QSPRpred/qsprpred/data/tables/pnds.py:443: Pandas4Warning: The 'verify_integrity' keyword in DataFrame.set_index is deprecated and will be removed in a future version. Directly check the result.index.is_unique instead.
    self.df.set_index(name, inplace=True, verify_integrity=True, drop=False)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 10 passed, 30 warnings in 6.19s =====
(QSPRPred_env) ..[manasmahale@Manas-MacBook-Pro] - [~/Desktop/QSPRpred/testing/test_pytest] - [Mon Jun 22, 08:02]
```


Conclusions

1. **Fixed the CLI Bug** ([#f85e767154e40472d64321c45c68f9a34f6a7fa9](#))
2. **Added PCAPlot** ([#9c06c890679ad2ada4ada9078b3653ab7d3d6894](#))
3. **Added UMAPPlot** ([#42195729e7e4971e7b0c20313c45d728527005e4](#))
4. **Added TSNEPlot** ([#68de560491a04c70a9527e957a28b6662e45ae8a](#))
5. **Added Tests for Above Plots** ([#8a8e96d97885eb238d9f3a4b6fd0e120c81dba85](#))
6. **Added Biased Splits and Tests** ([#ad252f65906347270cd4c4ba15f93fb765f4e288](#))